

# Murtuza Salman Mohammed

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## PROFESSIONAL SUMMARY

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Senior Software Engineer with **4+ years** of experience in **full-stack development** and **AI engineering**. Proven track record leading end-to-end project delivery and driving process improvements through scalable, reliable solutions and performance optimization. Skilled in **Java**, **Spring Boot**, and **React**, with a focus on building enterprise-grade applications for maintainability and high availability.

## PERSONAL OBJECTIVE

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Pursuing a responsible and challenging position in a growth-oriented organization where my qualification, knowledge, extensive technical experience, analytic and professional/personal skills can be better harnessed, shared, and enriched.

## PERSONAL SKILLS

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Innovative, Pro-Active Systematic Problem-Solving, Critical & Analytical Thinking, Ambitious & Goal-oriented, Handling Stressful Situations, Collaborative Teamwork

## CORE COMPETENCIES

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**Languages:** Java, Python, C#, JavaScript

**Frameworks & Libraries:** Spring Boot, React, Node.js, Express.js, TensorFlow, Scikit-learn, Keras, Apache Camel

**Cloud & Tools:** AWS (EC2, RDS, SageMaker), Azure, Docker, Redis, PostgreSQL, Azure Cosmos DB, Hibernate, JPA, RESTful APIs, Git

**Practices & Testing:** Agile, CI/CD, TDD, JUnit, Mockito, Spring Boot Test

**Monitoring & PM Tools:** Prometheus, Grafana, Datadog, Spring Boot Actuator, Jira, Trello, Swagger

## PROFESSIONAL EXPERIENCE

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**Senior Full Stack & AI Engineer** April 2025 – Present

Whiz Solutions Inc. – Calgary, AB, Canada

Developed scalable **backend**, **cloud**, and **AI/ML** solutions using **Java**, **Spring Boot**, **AWS**, **Azure**, **Docker**, and **Python**, with a focus on performance, security, and automation.

- Optimized backend performance through **SQL** query tuning, **Redis** caching, lazy loading, and efficient **JPA/Hibernate** data access, improving API response times by up to **25%**.
- Designed and deployed **cloud-native** applications and **microservices** using **Java**, **Spring Boot**, **Docker**, and **AWS** (EC2, RDS, SageMaker), supporting scalable, highly available production environments.
- Developed **RESTful APIs** and backend services using **Java**, **Spring Boot**, and **JPA/Hibernate**, focusing on performance, security, and scalability.
- Implemented enterprise integration and asynchronous communication patterns using **Apache Camel**.
- Integrated and optimized relational and **NoSQL** databases, including **PostgreSQL** and **Azure Cosmos DB**, applying indexing, partitioning, and caching techniques.

- Automated **CI/CD** workflows using **Azure DevOps**, **Jenkins**, **Git**, and **Docker**, contributing to a **30% reduction** in deployment-related defects.
- Implemented application observability using **Prometheus** and **Grafana**, improving visibility into API performance and infrastructure health.
- Built and deployed production **ML pipelines** on **Amazon SageMaker** with traffic shifting and automated rollback for reliable model deployments using **Guardrails**.
- Engineered a synthetic **computer-vision** data generation pipeline using **OpenCV** and **Albumentations**, producing **10,000+** annotated training images.
- Developed **reinforcement learning** solutions for autonomous systems using **AWS DeepRacer**, placing **1st among 20 teams** in an AI competition.
- Collaborated with engineers, data scientists, QA, and cloud architects to deliver end-to-end solutions across backend, cloud, and **AI/ML** deployment.

#### **Full-Stack Developer** *December 2021 – July 2023*

*Pilog India Pvt Ltd, Hyderabad, India.*

Developed and modernized **Java** applications and **APIs** for enterprise clients, improving performance, scalability, and security while mentoring developers and supporting **Agile** delivery.

- Designed and developed **Java**-based applications for finance and petrochemical clients.
- Migrated legacy codebase from **JDK 8** to **JDK 17**, improving performance by **40%**.
- Developed **RESTful APIs** using **Spring Boot**, integrating third-party systems and securing calls with **JWT**.
- Implemented load-balancing strategies for distributed applications, improving availability, and traffic distribution.
- Designed scalable database models and data access layers using **Hibernate** and **JPA**, applying transaction management and concurrency controls.
- Implemented pessimistic locking for critical financial transactions, maintaining data consistency and integrity.
- Managed database schema migrations using **Flyway**, automating version-controlled changes across environments.
- Mentored junior developers, conducted code reviews, and contributed to **Agile** adoption.
- Documented technical specifications, REST endpoints, and architecture diagrams to support maintainability and team handoffs.

#### **EDUCATION**

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##### **Bachelor of Engineering (B.E.), Computer Science** *(Graduated)*

*Southern Alberta Institute of Technology, Calgary, AB, Canada.*

##### **Artificial Intelligence with Machine Learning** *(Graduated)*

*Humber College, Toronto, ON, Canada.*

##### **Object-Oriented Software Development** *(Graduated)*

*Muffakham Jah College of Engineering and Technology, Hyderabad, India.*